

Agustín Avelino Pineda

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PROFESSIONAL SUMMARY

Mechatronics Engineering student specialized in embedded systems and power electronics. Led the electronics area of a 51.2V/300A competition kart and developed the full firmware for an autonomous AGV.

EDUCATION

Tecnológico de Monterrey Campus Santa Fe
B.S. in Mechatronics Engineering

Santa Fe, Mexico City
Expected graduation: August 2028

CECyT 9 “Juan de Dios Bátiz” — IPN
Technical Degree in Programming

Mexico City
March 2025

PROJECTS

Line-Following AGV with Load/Unload System
Engineering Project

Santa Fe, Mexico City
February - June 2026

- Designed and implemented a PD controller with dead zone and exponential smoothing filter on the PWM signal, achieving stable autonomous navigation on a differential-drive robot with BTS7960 drivers.
- Developed the complete control logic in C++: line following, path-loss recovery and station detection, iterating through 20+ firmware versions to reach a robust system.
- Integrated an HX711 load cell with a Kalman filter, encoders (494 PPR), a TMC2209 stepper motor for automated lifting, and a 400 kHz I2C architecture with PCF8574 and OLED on the same bus, achieving stable weighing in motion and repeatable load lifting.

LEADERSHIP AND ACTIVITIES

Omega Lightning — Student Racing Team, Tecnológico de Monterrey
Lead, Electronics Area

Santa Fe, Mexico City
February 2026 - August 2026

- Led the development of the electrical system of a 51.2V/300A competition kart, making component specification decisions and coordinating a team of 4 under competition deadlines.
- Performed component analysis and redesigned the kart wiring (from 3/0 to 6 AWG silicone cable), reducing system weight through proper selection of technical specifications.
- Designed (SolidWorks) and manufactured a sealed enclosure for component protection, applying design integrity and durability criteria, and implemented a Safe-to-Touch system to ensure electrical isolation of the chassis in high-power systems (51.2V/300A peak).
- Integrated solar panels for autonomous charging of the secondary electrical system (lighting, audio and electrical protection, independent from the motor power system) and developed a speed-adaptive audio system, improving pedestrian safety in the pit area.

Co-Lead, Electronics Area

August 2025 - December 2025

- Supported the design and integration of the kart electrical system, laying the groundwork for the project I later led.
- Installed the complete lighting system (headlights, taillights and brake lights) in a parallel arrangement, guaranteeing 100% lighting redundancy in case of failure.
- Collaborated on motor and battery integration and general technical development under competition deadlines.

SKILLS AND INTERESTS

Technical: C++, Python,
Git/GitHub, Linux, Arduino

Languages: Spanish (native),
English (Advanced)

Office: Excel (Intermediate),
Microsoft Office

Tools: SolidWorks, NX (Associate
Certification)

Interests: Robotics, AI, embedded
systems, motorsport engineering

Lab: Soldering, electronic systems
integration